

Feature Selection versus Feature Extraction Models for IoT Intrusion Detection System Using Convolutional Neural Network

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Abstract—Intrusion Detection Systems (IDS) are vital for securing Internet of Things (IoT) environments, which face escalating cyber threats due to the proliferation of interconnected devices. Traditional IDS methods that rely on signature-based detection fall short in addressing these evolving threats. This study aims to enhance IDS capabilities by evaluating feature selection and extraction techniques for multiclass label detection using Convolutional Neural Networks (CNNs). We preprocess the University of Nevada - Reno Intrusion Detection Dataset (UNR-IDD) through cleaning and normalization, followed by dimensionality reduction. Techniques such as Chi-Square, Lasso Regression, and Mutual Information were used for feature selection, while Principal Component Analysis (PCA), Factor Analysis, and Non-Negative Matrix Factorization (NMF) were used for feature extraction. Two CNN models, incorporating Conv1D and Conv2D layers, were trained and assessed. The findings reveal that Factor Analysis is the most effective feature extraction method, achieving accuracies of 86.1% with Conv1D and 87.6% with Conv2D CNNs, while Mutual Information excelled in feature selection, with accuracies of 78.5% for Conv1D and 79.2% for Conv2D CNNs. These results highlight the significant improvement in IDS performance when using advanced dimensionality reduction techniques. Future research should explore combining these techniques and applying them to diverse datasets to further enhance IDS accuracy and robustness across various IoT scenarios.

Keywords—Convolutional Neural Networks, Feature Extraction, Feature Selection, Internet of Things, Intrusion Detection Systems

I. INTRODUCTION

Intrusion Detection Systems (IDS) play a critical role in safeguarding Internet of Things (IoT) environments, which are becoming increasingly susceptible to cyber-attacks due to their rapid growth and interconnected nature [1]. The proliferation of IoT devices has significantly diversified and increased the complexity of network traffic, thereby expanding the potential attack surface and complicating the task of intrusion detection. Traditional IDS, which depend on signature-based detection methods, are no longer sufficient to address the dynamic and evolving nature of cyber threats. This situation has created a need for advanced methodologies, such as machine learning (ML) and deep learning (DL), to improve detection capabilities [2].

Research has demonstrated the effectiveness of ML and DL models in intrusion detection, employing various techniques for feature selection and extraction to enhance

model performance [3, 4]. Nevertheless, there remains uncertainty regarding the comparative effectiveness of feature selection versus feature extraction, particularly in multiclass label scenarios within IoT intrusion detection using Convolutional Neural Networks (CNNs) [5, 6]. The challenge of selecting the optimal technique that balances dimensionality reduction with high detection accuracy persists.

Recent research emphasizes the potential of integrating artificial intelligence (AI) and natural language processing (NLP) to enhance cybersecurity by improving threat detection and response. For instance, studies like [7] reveal that AI and NLP can significantly advance the identification of abnormal patterns and malicious intents within large datasets, which is essential for refining network security, minimizing false positives, and implementing adaptive security measures. Similarly, advancements in deep learning models for IDS, particularly in IoT environments, have been notable. [1] introduced the Deep Integrated Stacking for the IoT (DIS-IoT), an ensemble model that merges multiple DL models to achieve high accuracy and low false-positive rates. [3] further explored advanced feature extraction and selection techniques using CNNs and the Aquila optimizer, demonstrating performance competitive with traditional algorithms. [4] proposed a deep learning model tailored for Industrial Control Systems (ICSs), showing superior performance compared to conventional classifiers. These studies highlight the crucial role of advanced feature selection and extraction methods in improving IDS performance in complex IoT and ICS settings.

Additionally, research has shown diverse approaches to enhancing IoT intrusion detection with machine learning and neural networks. [8] investigated vulnerabilities in the MQTT protocol to Denial of Service (DoS) attacks and proposed a machine learning-based framework for accurate threat detection. [9] employed statistical methods such as Kullback-Leibler distance (KLD) and kernel density estimation to achieve high performance in detecting DoS and DDoS attacks. [10] developed a classification model that integrates heuristic features with various machine learning algorithms to improve phishing attack detection. At the same time, [11] proposed lightweight neural networks, including Conv1D CNNs, for effective detection with minimal computational requirements. These contributions collectively emphasize the importance of feature selection and preprocessing in creating effective IoT intrusion detection systems, showcasing a range of techniques to address modern cybersecurity challenges. This study aims to systematically compare feature selection and feature

extraction techniques to identify the most effective method for multiclass label detection in IoT environments using CNNs, thereby contributing to the development of more robust and accurate IDS capable of mitigating sophisticated and evolving threats in contemporary IoT infrastructures.

This paper begins with an introduction that provides a comprehensive overview of the utilization of Machine Learning in IoT data analysis, highlighting the importance of dimensionality reduction techniques and CNN models. The following section highlights an approach that utilizes various feature selection and extraction models and Convolutional Neural Networks (CNN). The results section outlines the evaluation process by presenting findings based on the accuracy of each algorithm and discusses their implications. The discussion interprets the results, highlighting the strengths and limitations of each method, and provides insights into selecting dimensionality reduction techniques based on evaluation metrics and data characteristics. Finally, the conclusion synthesizes the main findings, emphasizes the importance of this research in advancing IoT data analysis methods, and suggests future research directions for refining the analysis of IoT data.

II. RESEARCH METHODOLOGY

In this research, the University of Nevada - Reno Intrusion Detection Dataset (UNR-IDD) was employed [12]. This dataset improves upon existing ones by offering a broader range of samples and scenarios for better representation of network intrusions. It contains essential features for Machine Learning-based Network Intrusion Detection Systems (NIDS), such as Switch ID, Port Number, and traffic-related metrics like Received Packets and Sent Bytes. It also includes labels for various attack types like TCP-SYN Flood and Port Scan, enhancing the dataset's value for developing more accurate NIDS. After removing non-relevant columns like Switch ID and Port Number, the dataset was refined to 29 columns, making it more focused for analysis.

The Label column, which represents the type of network intrusion, was encoded using Label Encoding to convert categorical values into numerical format. The data was normalized using MinMaxScaler to prepare the features for model training, which scales the feature values to a range between 0 and 1. This normalization process helps to standardize the data, improving the performance and stability of machine learning models. After these preprocessing steps, the dataset was split into features (X) and labels (y). The features were normalized and converted back into a data frame for further analysis.

In this research, various feature selection and extraction techniques were evaluated to determine their impact on the performance of Conv1D and Conv2D Convolutional Neural Networks (CNNs) for intrusion detection. Feature selection methods, such as Chi-square, Lasso Regression, and Mutual Information, were used to identify and retain the most relevant features from the dataset. In contrast, feature extraction techniques, including PCA, Factor Analysis, and NMF, transformed the data into new feature representations to reduce dimensionality. The processed data from these techniques was then used to train and assess not only the effectiveness of both Conv1D and Conv2D CNN models in detecting network intrusions but also the running time of each model.

The features initially in a Conv1D format were reshaped into Conv2D matrices to adapt the data for processing with Conv2D CNNs. The process involved several steps. First, the data was split into training and testing sets. Next, the number of features in the dataset was determined to decide on an appropriate matrix size. For instance, if the number of features was less than or equal to 64, the data was reshaped into an 8x8 matrix; if the number of features was less than or equal to 256, a 16x16 matrix was used. Padding was applied to adjust the number of features to fit the chosen matrix size. The data was then reshaped into 4D arrays with the shape (samples, height, width, channels), where height and width are the dimensions of the matrix and channels represent the number of data channels, typically set to 1 for grayscale images. This reshaped data was then used to train the Conv2D CNN model, which processes the data in a format suitable for Conv2D operations. These steps are illustrated in Fig. 1. Meanwhile Fig. 2 showed distribution of label in dataset used in this research.

A. Feature Selection

1) Chi-square

The Chi-Square method is a popular feature selection technique that evaluates the independence of each feature from its category to eliminate irrelevant factors before classification [13]. This filter-based approach determines feature importance by assessing the dependence of a class on a given feature. The Chi-Square algorithm operates by selecting features based on their Chi-Square values (X^2) and merging adjacent interval pairs with the lowest X^2 values, as long as these values exceed a significance threshold. The Chi-Square statistic is calculated using (1) and (2) [14].

$$X^2 = \sum_{i=1}^2 \sum_{j=1}^k \frac{(A_{ij} - E_{ij})^2}{E_{ij}} \quad (1)$$

$$E_{ij} = \frac{A_i \times A_j}{N} \quad (2)$$

where A_{ij} represents the observed frequency, E_{ij} is the expected frequency, k is the number of classes, A_i is the number of patterns in the i -th interval, A_j is the number of patterns in the j -th class, and N is the total number of patterns. This method's simplicity and effectiveness make it widely used for feature selection in machines.

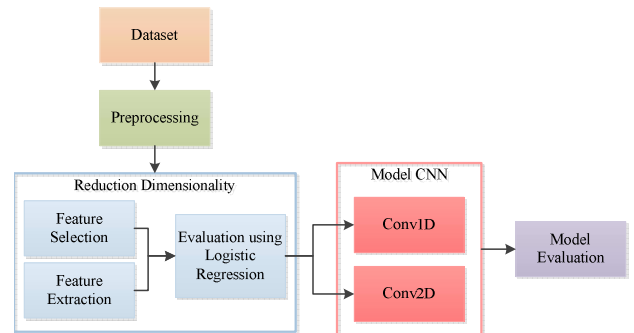


Fig. 1. Research Methodology

1) Lasso regression

Lasso regression feature selection is a powerful technique used in machine learning to enhance model performance by identifying and selecting the most important features while reducing the impact of irrelevant ones. Lasso, which stands for Least Absolute Shrinkage and Selection Operator, applies an L1 penalty to the regression coefficients during model training. This penalty forces some of the coefficients to shrink towards zero, effectively performing feature selection by excluding features with coefficients that are reduced to zero. The main advantage of Lasso regression is its ability to handle high-dimensional data and prevent overfitting by promoting sparsity in the feature space [15]. As a result, Lasso helps in improving the interpretability of the model and enhances its generalization capability by retaining only the most significant predictors. Lasso regression is calculated using (3) [15].

$$\hat{\beta}_{LASSO} = \underset{\beta}{\operatorname{argmin}} \left[-1(\beta, y_j) + \lambda \sum_{j=1}^f |\beta_j| \right] \quad (3)$$

Where f is the reduced number of features, λ is the tuning parameter that controls the strength of the lasso penalty.

2) Mutual Information

Mutual information (MI) feature selection is a widely utilized filter method for enhancing the performance of intrusion detection systems (IDS) in various Internet of Things (IoT) environments. This technique evaluates the dependency between each feature and the class label, selecting features based on their mutual information scores. The primary advantage of mutual information is its ability to capture nonlinear relationships between features and class labels, which is crucial for handling complex and nonlinear data patterns typical in IoT environments. Additionally, mutual information is effective in dealing with noisy and incomplete data, expected in IoT networks, thereby improving the accuracy and reliability of IDS by reducing data dimensionality and retaining the most relevant features. Mutual information is calculated using (4) [16].

$$MI(X; Y) = H(X) - H(X|Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \quad (4)$$

Where $H(X)$ is the entropy of X , $H(X|Y)$ is the conditional entropy of X given Y , $p(x)$ and $p(y)$ are the marginal distributions of x and y , respectively. The entropy $H(X)$ is given by (5) and the conditional entropy $H(X|Y)$ is given by (6).

$$H(X) = - \sum_{x_i \in X} p(x_i) \log p(x_i) \quad (5)$$

$$H(X|Y) = - \sum_{y_j \in Y} p(y_j) \sum_{x_i \in X} p(x_i|y_j) \log p(x_i|y_j) \quad (6)$$

By selecting features with the highest mutual information scores, this method effectively identifies the most informative features for IDS, enhancing system performance and robustness against attacks [16].

B. Feature Extraction

1) Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms the original data into a set of orthogonal vectors, called principal components, to capture the maximum variance and reduce redundancy. The process involves computing the covariance matrix C of the centred data, which is calculated using (7) [17].

$$\operatorname{cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad (7)$$

Where n is the number of samples, the covariance matrix is then decomposed into eigenvalues Λ and eigenvectors U , forming the principal component matrix as calculated as given by (8).

$$C_p^{*p} = U_p^{*p} \Lambda_p^{*p} U_p^{*p} \quad (8)$$

The original data is projected onto the new coordinate system using (9).

$$F_{n \times p} = X_{n \times p} U_{n \times p} \quad (9)$$

The cumulative contribution rate of each principal component is calculated to determine the proportion of variance explained by (10).

$$\operatorname{pre}_k = \frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^p \lambda_i} \quad (10)$$

The calculation helps in selecting a reduced number of components d , where $d < p$, to retain essential information for data analysis.

2) Factor Analysis

Factor Analysis is a sophisticated multivariate statistical method designed to uncover the underlying structure within large datasets by identifying latent factors that influence variability among observed variables. It aims to elucidate the essential factors that explain the correlation patterns between variables, thus revealing the fundamental relationships and reducing the complexity of the data [18]. In essence, Factor Analysis helps to filter out meaningless or redundant attributes by focusing on meaningful variables that capture the essence of the data [19]. This method is particularly effective in large-scale analyses, where traditional techniques may fall short, by isolating the significant attributes from a multitude of observed variables and thereby enhancing the clarity and interpretability of the dataset.

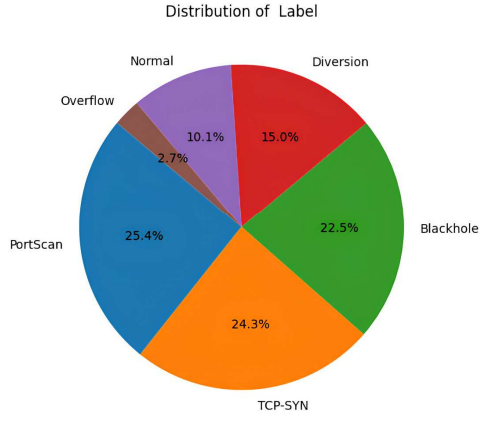


Fig. 2. The distribution of each label in the dataset

3) Non-Negative Matrix Factorization (NMF)

Non-negative Matrix Factorization (NMF) is a powerful technique for feature extraction and dimensionality reduction. NMF ensures that features are non-negative by generating additive models, which is advantageous for applications like text mining, where the same word can have different meanings depending on its context [20]. Furthermore, in the field of bioinformatics, NMF is employed to identify 'metagenes' from expression profiles, revealing associations with biological pathways. NMF involves finding matrices W and H such as given in (11).

$$A \approx WH \quad (11)$$

Where A is an $m \times n$ non-negative matrix, W is an $m \times k$ non-negative matrix, and H is a $k \times n$ non-negative matrix. The optimization aims to minimize the difference between A and WH , often using a cost function like the Frobenius norm as given in (12).

$$\min_{W,H} \|A - WH\|_F \quad (12)$$

This method optimizes data representation by minimizing the norm of the difference $A - WH$, where A is the original non-negative matrix and W and H are the reduced feature and importance matrices, respectively [21].

C. CNN Model

Convolutional Neural Networks (CNNs) are highly effective in capturing spatial and structural patterns within data, making them particularly suitable for image analysis and feature extraction. Specifically, in the context of intrusion detection, CNNs can analyze network traffic data and packet payloads [22]. They excel in recognizing spatial relationships and patterns that may indicate malicious activity, which traditional methods might overlook.

Moreover, CNNs are primarily utilized in computer vision because they mimic the human visual system. Comprising

multiple convolutional layers that operate on Conv2D matrices, these networks are instrumental in feature extraction, enabling the effective processing and understanding of images [23]. The main distinction between Conv1D and Conv2D CNNs lies in their dimensionality and application: Conv1D CNNs process one-dimensional data such as time series or sequential data, capturing patterns along a single axis, whereas Conv2D CNNs handle two-dimensional data like images, capturing patterns across both height and width dimensions. This distinction is crucial for tasks involving spatial relationships within images—architecture of Conv2D CNN model as shown in Fig. 3. Furthermore, Table I shows the CNN model architectures used in this research.

D. Evaluation

Evaluation of a machine learning model often includes various metrics, with accuracy being one of the most commonly used. Accuracy measures the proportion of correctly predicted instances out of the total instances in the dataset. Besides that, in this research Precision, Recall, F1Score and running time of the models is also calculated. These measures is used to compared each model to get more comprehensives result about its performances.

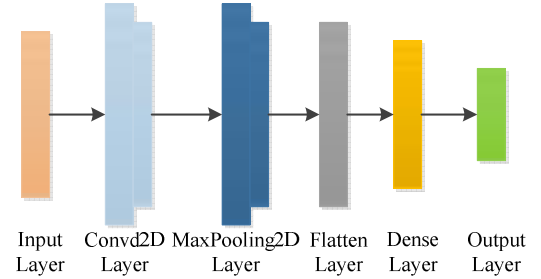


Fig. 3. Architecture of Conv2D CNN Model

TABLE I. CNN MODEL ARCHITECTURES

Parameter	Conv2D CNN	Conv1D CNN
Convolution Layer	64 filters, Kernel size: (3, 3), Input shape: (height, width, 1)	Conv1D: 64 filters, Kernel size: 2, Input shape: (X_train.shape[1], 1)
Pooling Layer	MaxPooling2D: Pool size: (2, 2)	MaxPooling1D: Pool size: 2
Flatten Layer	Flatten: Flattening the 2D matrices into a vector	Flatten: Flattening the 1D matrices into a vector
Dense Layer	Dense: 100 units	Dense: 100 units
Output Layer	Dense: 6 units	Dense: 6 units
Activation	Relu, Softmax	Relu, Softmax
Optimization Algorithm	Adam	Adam

TABLE II. RESULT OF THE COMBINATION OF FEATURE SELECTION AND FEATURE EXTRACTION MODELS USING CONV1D AND CONV2D CNN

		# Features	Conv1D CNN					Conv2D CNN				
			Precision	Recall	F1Score	Accuracy	Time	Precision	Recall	F1Score	Accuracy	Time
Features selection models	Chi-square	19	0.790	0.780	0.770	0.777	36.901	0.710	0.700	0.690	0.704	24.780
	Lasso Regression	4	0.720	0.740	0.720	0.741	26.860	0.740	0.750	0.730	0.750	23.945
	Mutual Information	14	0.790	0.780	0.780	0.785	30.909	0.800	0.790	0.790	0.792	69.525
Features extraction models	PCA	14	0.780	0.760	0.750	0.760	31.574	0.780	0.770	0.770	0.772	77.821
	Factor Analysis	14	0.880	0.860	0.860	0.861	53.882	0.880	0.880	0.870	0.876	95.627
	NMF	4	0.680	0.690	0.680	0.693	30.162	0.720	0.710	0.690	0.711	88.865

TABLE III. STATISTIC DESCRIPTIVE FOR THE RESULT

CNN Models	Evaluation	Count	Mean	Standar deviation	Min	25%	50%	75%	Max
Conv1D CNN	Precision	6	0.770	0.056	0.693	0.746	0.769	0.783	0.861
	Recall	6	0.760	0.061	0.680	0.728	0.760	0.778	0.860
	F1Score	6	0.773	0.069	0.680	0.735	0.785	0.790	0.880
	Accuracy	6	0.768	0.056	0.690	0.745	0.770	0.780	0.860
	Time	6	35.048	9.780	26.860	30.349	31.242	35.569	53.882
Conv2D CNN	Precision	6	0.768	0.063	0.704	0.721	0.761	0.787	0.876
	Recall	6	0.757	0.069	0.690	0.700	0.750	0.785	0.870
	F1Score	6	0.772	0.063	0.710	0.725	0.760	0.795	0.880
	Accuracy	6	0.767	0.065	0.700	0.720	0.760	0.785	0.880
	Time	6	63.427	31.562	23.945	35.966	73.673	86.104	95.627

III. RESULT

The study investigated the performance of various dimensionality reduction techniques applied to Conv1D and Conv2D Convolutional Neural Networks (CNNs) for intrusion detection. The evaluation included both feature selection and feature extraction models, and the comparison results as showed in Table II.

Among the feature selection models, Mutual Information demonstrated the highest accuracy for both CNN architectures. Specifically, it achieved an accuracy of 0.785 with Conv1D CNN and 0.792 with Conv2D CNN. This indicates that Mutual Information effectively retained essential features, leading to superior model performance compared to other methods. Conversely, Chi-square and Lasso Regression exhibited lower accuracy. Chi-square achieved an accuracy of 0.777 with Conv1D and 0.704 with Conv2D, while Lasso Regression recorded accuracies of 0.741 and 0.750 for Conv1D and Conv2D, respectively.

In the realm of feature extraction models, Factor Analysis stood out with the highest accuracy across both Conv1D and Conv2D CNNs. For Conv1D CNN, Factor Analysis achieved an accuracy of 0.861, and for Conv2D CNN, it reached 0.876. This result underscores the effectiveness of Factor Analysis in enhancing model accuracy by effectively capturing and utilizing critical features. On the other hand, Non-Negative Matrix Factorization (NMF) showed the lowest performance, with accuracies of 0.693 for Conv1D and 0.711 for Conv2D, reflecting its limited capacity in improving the model's performance. The statistical analysis presented in Table III

provides additional insights into the performance metrics for Conv1D and Conv2D CNN models. For Conv1D CNN, the mean accuracy was 0.768 with a standard deviation of 0.056, suggesting moderate variability in performance across different techniques. The precision, recall, and F1Score metrics had mean values of 0.770, 0.760, and 0.773, respectively, with a range of 0.693 to 0.861 in precision. This variability highlights the differences in effectiveness among the dimensionality reduction techniques.

For Conv2D CNN, the mean accuracy was 0.767 with a standard deviation of 0.065, indicating a similar level of variability. The precision, recall, and F1Score metrics had mean values of 0.768, 0.757, and 0.772, respectively, with accuracy ranging from 0.700 to 0.880. The higher maximum values for both precision and accuracy in Conv2D CNN reflect the impact of different dimensionality reduction techniques on model performance. Figures 4 and 5 illustrate the training and validation loss and accuracy curves for Mutual Information with Conv1D CNN and Factor Analysis with Conv2D CNN, respectively.

The results show that Mutual Information and Factor Analysis are the top dimensionality reduction techniques for improving CNN-based intrusion detection systems. Mutual Information was best for feature selection, while Factor Analysis was the most effective for feature extraction. The statistical analysis supports these findings, highlighting the need to select the right dimensionality reduction methods to enhance CNN performance in intrusion detection.

IV. DISCUSSION

In the comparative analysis of Intrusion Detection System (IDS) models, the current study offers a focused examination of feature selection and extraction techniques specifically applied to Convolutional Neural Networks (CNNs). This approach diverges from previous research that predominantly explored a variety of traditional and advanced models without a specific focus on dimensionality reduction methods within the CNN context. Previous studies, such as those by [24] and [25], employed conventional feature extraction methods like Principal Component Analysis (PCA) and Analysis of Variance (ANOVA). These methods achieved accuracies of up to 79.5% with Multi-Layer Perceptron (MLP) models and up to 99% with Fully Convolutional Networks (FCN) and Random Forest methods. In contrast, this study's focus on Mutual Information and Factor Analysis presents a more precise approach, demonstrating that these techniques can achieve even higher accuracy, with up to 87.6% accuracy in Conv2D CNNs.

Furthermore, while [12] reported high performance with various classification methods, including Random Forest and MLP, it did not specifically address the impact of dimensionality reduction techniques. This study's thorough examination of feature selection and extraction methods underlines their critical role in enhancing IDS performance, contributing significantly to the field.

Overall, this study emphasizes the efficacy of Mutual Information and Factor Analysis in advancing CNN-based intrusion detection systems. The comparative analysis not only confirms the superior performance of these techniques but also highlights their advantages over traditional methods, offering valuable insights for improving IDS models in IoT

environments. A detailed comparison of these methodologies and results can be seen in Table IV, which refers to the comparative table included in the study.

V. CONCLUSION

This study aimed to systematically compare feature selection and extraction techniques to identify the most effective method for multiclass label detection in IoT environments using CNNs. The results demonstrate that Mutual Information and Factor Analysis are the most effective dimensionality reduction techniques for enhancing CNN-based intrusion detection systems. Mutual Information achieved the highest accuracy with Conv1D and Conv2D CNNs, reaching 78.5% and 79.2% respectively, while Factor Analysis excelled in feature extraction, attaining accuracies of 86.1% for Conv1D and 87.6% for Conv2D CNNs. These findings underscore the importance of selecting the appropriate dimensionality reduction techniques to optimize CNN model performance in detecting threats within complex and evolving IoT environments.

The study demonstrates the advantages of Mutual Information and Factor Analysis compared to traditional techniques, such as PCA and ANOVA, which have achieved maximum accuracies of 79.5% with Multi-Layer Perceptron models and up to 99% with Fully Convolutional Networks. This indicates a significant opportunity for improving accuracy through advanced dimensionality reduction techniques. To further develop more effective intrusion detection systems, future research should investigate various combinations of these advanced methods and apply them to a broader range of datasets. This approach will help to better understand and optimize their effectiveness in enhancing IDS performance across diverse IoT environments.

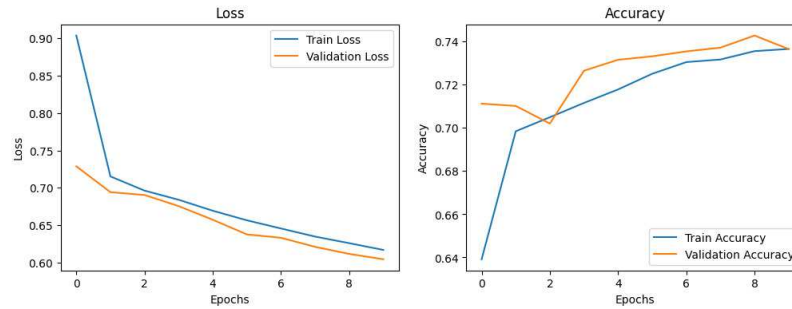


Fig. 4. Training and Validation Loss and Accuracy Curves in Lasso Regression using Conv1D CNN Architecture

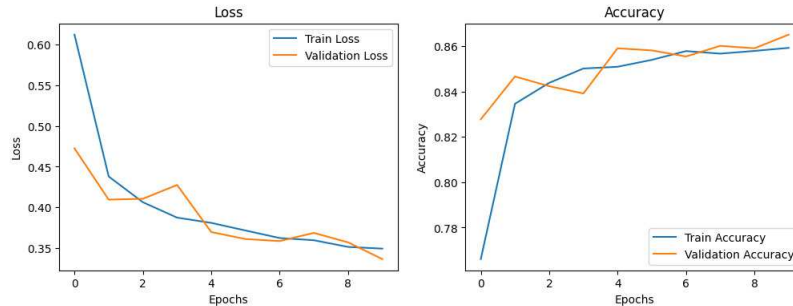


Fig. 5. Training and Validation Loss and Accuracy Curves in Factor Analysis using Conv2D CNN Architecture

TABLE IV. COMPARISON RESEARCH USING THE UND-IRR DATASET

Paper	Feature Extraction	Classification Method	Result
[24]	PCA and ANOVA	Multi-Layer Perceptron (MLP) with eight hidden layers	Baseline Model: Training accuracy: 77.22%, Test accuracy: 71.87% Final Tuned Model: Training accuracy: 80.16%, Test accuracy: 79.5%
[25]	Random Forest	Fully Convolutional Networks (FCN) and MLP	Accuracy: 99%, Precision: 100%, Recall: 99%, F1 Score: 99%, AUC: 1.0
[12]	-	Random Forest, MLP, SVM, Bagging Classifier, K-Neighbors Classifier, AdaBoost Classifier	Binary Classification Performance: Precision, Recall, F1 Score: 1.0 Multiclass Classification Performance: RF, BC, SVM, MLP, KNC, ABC performance metrics
Current research	Feature Selection: Chi-Square, Lasso Regression, Mutual Information Feature Extraction: PCA, Factor Analysis, NMF Descriptive Analysis	Conv1D CNN Conv2D CNN	Mutual Information: Conv1D Precision: 0.790, Recall: 0.780, F1Score: 0.780, Accuracy: 0.785, Conv2D Precision: 0.800, Recall: 0.790, F1Score: 0.790, Accuracy: 0.792 Factor Analysis: Conv1D Precision: 0.880, Recall: 0.860, F1Score: 0.860, Accuracy: 0.861, Conv2D Precision: 0.880, Recall: 0.880, F1Score: 0.870, Accuracy: 0.876

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